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Localization of marine pollutant sources: Gradient Method vs Reinforcement Learning (PPO)

Velocity modulation and a dual sensor formation

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Contents

1	Introduction	2
2	Uniform Spawn in $[d_{\min}, d_{\max}]$	2
2.1	Definition of the spawn disk	2
2.2	The 300 m floor is a weak constraint	3
2.3	Uniform sampling by distance	4
2.4	Visualization: spawn maps	4
3	Adaptive FCM vs. Adaptive PPO	7
3.1	Adaptive FCM (Adam)	7
3.2	Adaptive PPO (variable-velocity chain)	8
3.3	Quantitative FCM–PPO comparison	13
4	Modulation Test: 10 m Success Radius	16
4.1	Motivation	16
4.2	Setup	16
4.3	Results	17
5	Single vs. Double Ring Formation	19
5.1	Motivation and comparison protocol	19
5.2	Observation architecture	21
5.3	Results: comparison at each $v_{\max}$	21
5.4	Where the double ring loses: differential failure analysis	22
5.5	Interpretation	25
5.6	Agent occupancy heatmaps	25
6	Conclusions	27

1 Introduction

This report starts *in medias res*: it does not introduce the problem from scratch, but builds on results already obtained previously. The task is *source seeking* — an autonomous marine agent must reach the source of a pollutant starting from a point inside the plume, guided only by local concentration measurements, in a 2D marine domain (Cecina bay, 3×2.5 km, 10 m resolution, 10 s time step) with wind and currents. An episode is a **success** if the agent enters within a radius of the source; the metrics are the **success rate** (SR, fraction of successful episodes) and the number of **steps to success** (speed).

The two baselines **already exist**, both at **constant velocity**:

- **Pure FCM** — the gradient method (*Field Climbing Method*), *training-free* and fixed-step; in its best configuration (sensor range 50 m) it reaches an SR of **32.9%**;
- **PPO minimal** — a policy learned with reinforcement learning at fixed velocity (1 m/s), much more reliable: **96.9%**.

The ~ 64 percentage-point gap between the two quantifies how much a learned policy — with episode memory and hydrodynamic context — outperforms pure local gradient ascent.

On this basis, the report first introduces a **new spawn metric**: the episode start points are distributed *uniformly by distance* in a disk $[d_{\min}, d_{\max}]$ adapted to the plume’s diffusion, replacing the previous fixed-interval distribution. All the evaluations that follow are produced with this metric. It then studies the transition to **variable velocity**, comparing its two versions — FCM made adaptive with Adam and the PPO chain at increasing velocity — with two in-depth analyses: a velocity **modulation** test (success radius reduced to 10 m) and the addition of a **second sensor formation** (double ring). The evaluation uses 26 test sources (SRC107–132) over four wind scenarios (V0–V3) and three time windows (Q1/4, Q1/2, Q3/4), described below.

2 Uniform Spawn in $[d_{\min}, d_{\max}]$

2.1 Definition of the spawn disk

For each scenario (source, wind, time chunk) the spawn point $\mathbf{p}$ is chosen so as to satisfy a set of **constraints** that define an *admissible disk* $[d_{\min}, d_{\max}]$ centered on the source $\mathbf{p}_s$, where $d = \|\mathbf{p} - \mathbf{p}_s\|$ is the distance from the source. Letting $\{d_i\}$ be the set of distances of the grid cells belonging to the plume, the constraints are:

$$\begin{aligned}
 \text{(V) validity: } & C(\mathbf{p}) \geq C_{\text{thr}}, \quad \text{dist}_{\text{land}}(\mathbf{p}) \geq \ell_{\min} \\
 \text{(E) outer edge: } & d_{\max} = \min \left(d_{\text{cap}}, \max_i d_i \right) \\
 \text{(I) inner edge: } & d_{\min} = \max \left(d_{\text{floor}}, P_{50}(\{d_i\}) \right) \\
 \text{(W) minimum width: } & d_{\max} - d_{\min} \geq w_{\min}
 \end{aligned}$$

with $d_{\text{cap}} = 3000$ m, $d_{\text{floor}} = 300$ m, $w_{\text{min}} = 500$ m. Within the disk defined this way, the spawn point is **sampled** in three steps:

- (U) target distance: $d^* \sim \mathcal{U}(d_{\text{min}}, d_{\text{max}})$
- (B) cell selection: $i^* \in \{i : |d_i - d^*| \leq \delta_b\}$
- (J) jitter: $\mathbf{p} = \mathbf{c}_{i^*} + (\varepsilon_x, \varepsilon_y), \quad \varepsilon_x, \varepsilon_y \sim \mathcal{U}(-\delta_j, \delta_j)$

with $\delta_b = 50$ m and $\delta_j = 5$ m ($\mathbf{c}_{i^*}$ is the center of the selected cell). Each element has a precise role:

- **(V) Validity** — the spawn must fall on a cell *inside the plume* (concentration above threshold C_{thr}) and at a safe distance from land (ℓ_{min}). It is the **strong**, non-negotiable constraint: the others are subordinate to it.
- **(E) Outer edge** — the maximum radius is the smaller of an absolute cap $d_{\text{cap}} = 3$ km and the farthest plume cell. It prevents spawns beyond 3 km even when the plume extends farther.
- **(I) Inner edge** — the minimum radius is the larger of a floor $d_{\text{floor}} = 300$ m and the **median** P_{50} of the plume distances. Using the median makes d_{min} **scale with diffusion**: extended plume $\Rightarrow$ high median $\Rightarrow$ high d_{min} (far spawns); compact plume $\Rightarrow$ low d_{min} (near spawns). The floor *tends to* avoid spawns in the source's immediate radius, but it is a *weak* constraint (next subsection).
- **(W) Minimum width** — the interval must be at least $w_{\text{min}} = 500$ m wide; if it turns out too narrow, d_{min} is lowered (down to the floor) to ensure a non-degenerate spawn band. In the limiting cases of a very compact plume, where even this is not enough, the safety clause $d_{\text{min}} = \min_i d_i$ kicks in (sampling over the plume's actual extent; cf. next subsection).
- **(U) Target distance** — a distance d^* is drawn *uniformly* in $[d_{\text{min}}, d_{\text{max}}]$. It is this step that makes the sampling uniform *by distance* and not by area (clarified in §2.3).
- **(B) Cell selection** — a valid plume cell is chosen at random whose distance from the source differs from d^* by less than $\delta_b = 50$ m (if none falls in the band, the closest to d^* is taken).
- **(J) Jitter** — a small uniform noise $\pm\delta_j = \pm 5$ m is added to the cell position on each coordinate, so as not to always spawn exactly at the cell center.

2.2 The 300 m floor is a weak constraint

The **strong** constraint is that the spawn falls on a valid plume cell; the floor $d_{\text{floor}} = 300$ m is instead a *weak constraint* and can be violated. In practice a fraction of the spawns end up below 300 m through two distinct mechanisms.

(a) Sampling-band tolerance. Even with $d_{\text{min}} = 300$ m, the sampling accepts a cell within $|d_i - d^*| \leq 50$ m of the target (plus ± 5 m jitter): with a target close to d_{min} , cells down to ~ 250 m are admitted. These are the spawns just below threshold.

(b) Compact-plume fallback. The 300 m floor presupposes that plume cells exist beyond that distance. When the whole plume is closer (e.g. poorly diffused scenarios where all cells lie within ~ 200 m), one has $d_{\text{max}} = \max_i d_i < 300$ m and the floor

cannot be honored; the safety clause $d_{\min} = \min_i d_i$ then kicks in, and sampling is done over the plume’s **actual extent**. In these scenarios a spawn at 300 m simply does not exist among the valid cells.

2.3 Uniform sampling by distance

Step (U) draws the target distance *uniformly over distance*, not over the disk’s **area** — and this is a deliberate choice. Uniform sampling over area would concentrate spawns on the **outer ring**: at distance r the available surface grows with r (a thin ring has area $\approx 2\pi r dr$), so most points would fall near the outer edge. By instead drawing $d^* \sim \mathcal{U}(d_{\min}, d_{\max})$ and then choosing a cell at that distance (step B), **every distance band is equiprobable**, making training and evaluation balanced with respect to the initial distance.

On the evaluation data (1560 episodes), about 260 spawns fall below 300 m: about 100 in the 250–300 m band (band tolerance, mechanism *a*) and about 160 below 250 m (compact plumes, mechanism *b*, down to a few meters). The uniform coverage by distance therefore holds within the *effective* interval $[d_{\min}, d_{\max}]$ allowed by each scenario’s geometry.

2.4 Visualization: spawn maps

The *spawn maps* show, for the wind/chunk combinations with $\text{SR} < 100\%$, the start points colored by outcome (• success, ✕ failure) over the plume background, with the two dashed circles of the disk $[d_{\min}, d_{\max}]$ centered on the source. The distributions shown depend **only** on the spawn logic (not on the model used as colorer): the uniform coverage by distance is evident in the disk section between the two circles.

Spawn map SRC114

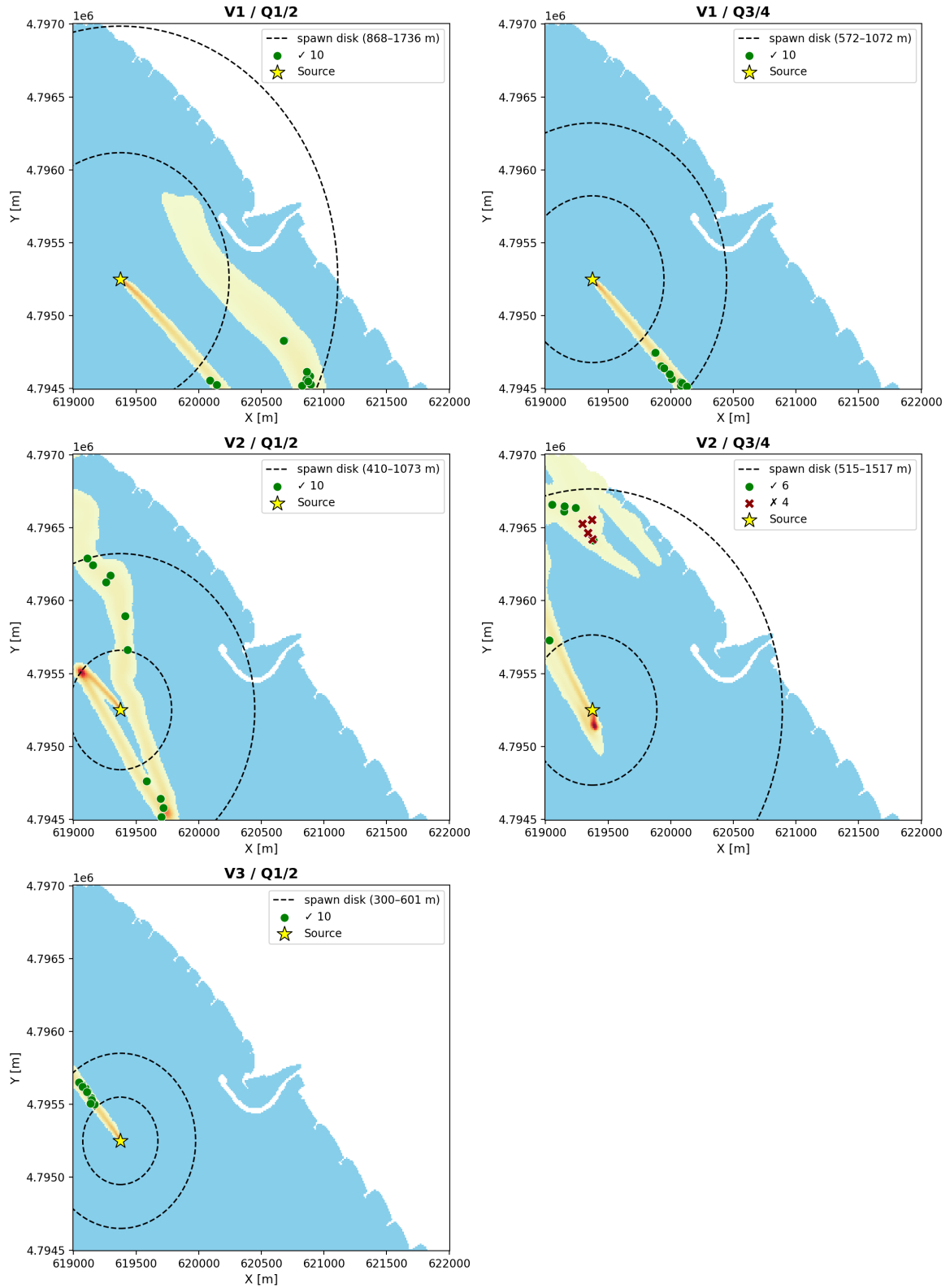


Figure 1: Spawn map — SRC114 (colorer: PPO single ring). Points distributed uniformly by distance in the disk $[d_{\min}, d_{\max}]$ (dashed circles, clipped to the domain).

Spawn map SRC132

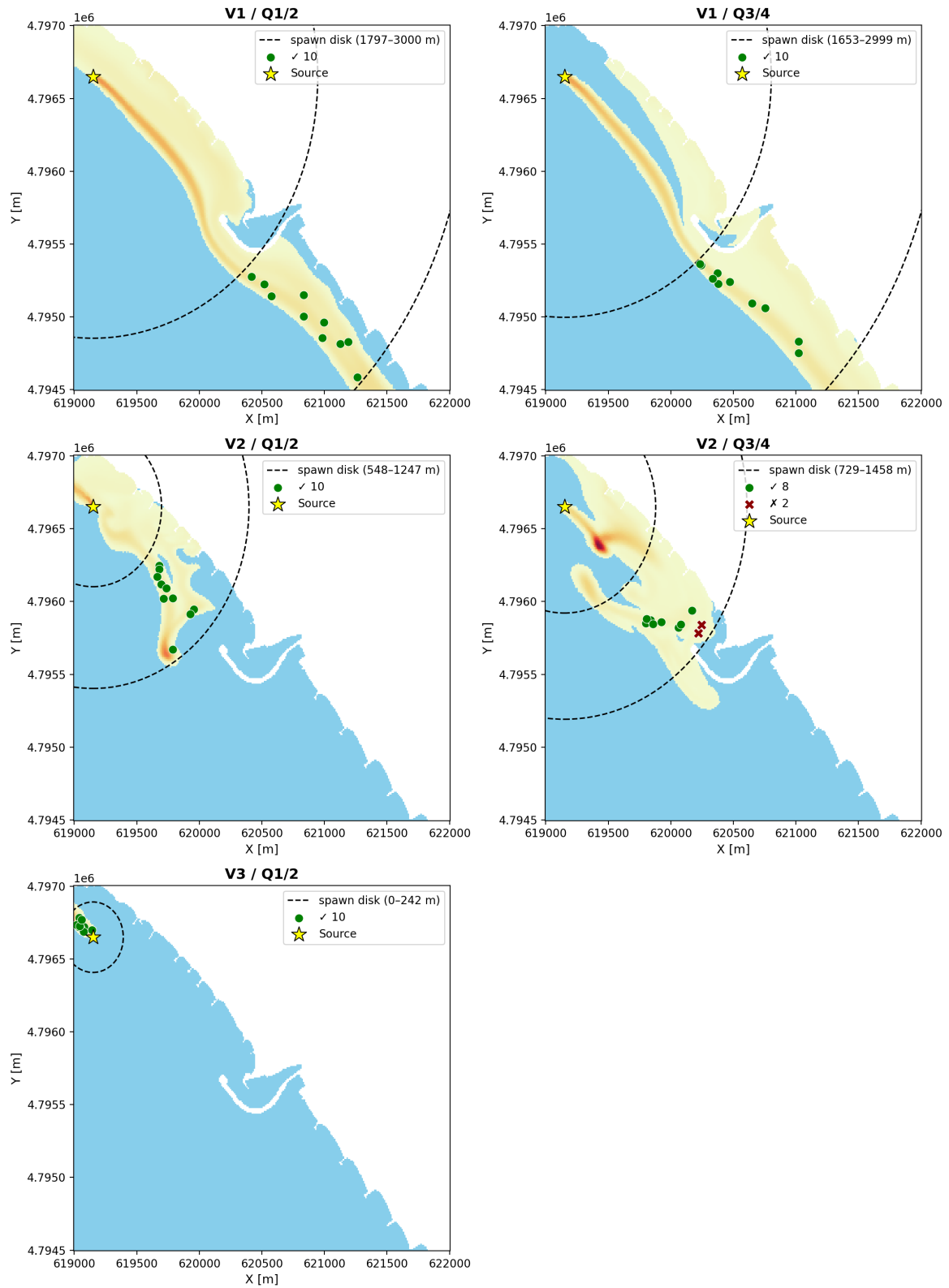


Figure 2: Spawn map — SRC132. The spawn band scales with each scenario's plume diffusion; successes and failures show no systematic spatial separation.

3 Adaptive FCM vs. Adaptive PPO

In the *source seeking* problem the agent must reach the source of a pollutant starting from a point inside the plume, having only local concentration measurements. A decisive factor for performance is the ability to **modulate the step**, i.e. the distance travelled at each timestep. A large step advances quickly but risks overshooting the source or being dragged by the current; a short step is more precise but slow and vulnerable to timeouts. Compared with a fixed-step agent, an **adaptive-step** agent can in principle speed up when far away and slow down near the source or against the current.

This chapter compares two opposite ways of achieving such modulation:

- **Adaptive FCM (Adam)** — a *training-free* method: the step is computed instant by instant by a deterministic rule applied to the locally estimated concentration gradient. It requires no training and has no memory beyond the gradient moments.
- **Adaptive PPO** — a policy *learned* with reinforcement learning: at each step it chooses the (direction, velocity) pair maximizing the expected return, exploiting the history of observations and the context variables (wind, current).

The comparison is made fair by putting the two methods at **equal maximum velocity**: each FCM lr value corresponds to an effective velocity comparable with the $v_{\max}$ of a PPO chain stage. The goal is to quantify how close a gradient climber, however refined by momentum, comes to a policy that has *learned* to exploit memory and context.

3.1 Adaptive FCM (Adam)

The *Field Climbing Method* (FCM) is a gradient climber: at each timestep it estimates the direction of maximum concentration increase and moves along it. The classic version uses a fixed step and reacts only to the current sensor reading, producing oscillating trajectories when the signal is noisy. The *adaptive* variant replaces the fixed-step rule with the **Adam** optimizer, which introduces a form of short-term memory on the gradient.

Gradient estimation. At each instant the agent has 8 directional sensors on a ring of radius $r_s = 50$ m. From these 8 readings it reconstructs the local gradient $\hat{\nabla}C \in \mathbb{R}^2$ via least-squares regression (first-order Taylor approximation of the concentration field around the current position).

Step with Adam. The estimated gradient is not followed directly, but feeds the Adam update, which maintains two moving averages: the *momentum* $\mathbf{m}$ (gradient average, $\beta_1=0.9$) and the *variance* $\mathbf{v}$ (square average, $\beta_2=0.999$). The step vector is $\mathbf{s} = lr \cdot \hat{\mathbf{m}}/(\sqrt{\hat{\mathbf{v}}} + \varepsilon)$. Two useful effects follow: the momentum **damps the oscillations** from sensor noise, keeping the direction coherent over several timesteps; the normalization by $\sqrt{\hat{\mathbf{v}}}$ **reduces the step** in the directions where the gradient is incoherent over time. The effective discrete direction is the one of the 8 axes most aligned with $\mathbf{s}$, and the step magnitude is capped at $\Delta x_{\max} = 50$ m.

Meaning of lr . The *learning rate* is the base step in meters: at convergence of the moments, for cardinal gradients one has $\|\mathbf{s}\| \rightarrow lr$. With $\Delta t = 10$ s this corresponds to

an effective velocity $lr/\Delta t$ (e.g. $lr=10\text{ m} \rightarrow 1\text{ m/s}$, $lr=50\text{ m} \rightarrow 5\text{ m/s}$). High lr values exceed the vehicle’s nominal velocity (1 m/s) and are therefore physically unrealistic: they are included in the sweep only for comparability with the PPO chain at increasing $v_{\max}$. The sweep is run on 624 episodes per value ($26\text{ sources} \times 4\text{ winds} \times 3\text{ chunks} \times 2\text{ ep}$), with the new spawn metric.

The tables that follow report the success rate also disaggregated by **wind scenario** and **time window**, the same two axes used throughout the report: **V0–V3** are the four wind/current conditions (V2 is the most adverse, sustained current); **Q1/4**, **Q1/2**, **Q3/4** are three instants of the simulation (at 1/4, 1/2 and 3/4 of its duration), which fix the plume state at the start of the episode.

lr (vel.)	SR%	Step (min)	V0	V1	V2	V3	Q1/4	Q1/2	Q3/4
10 m (1 m/s)	54.6	174.0 (29.0)	73.7	29.5	35.9	79.5	77.9	30.3	55.8
20 m (2 m/s)	58.2	131.5 (21.9)	80.1	32.1	34.6	85.9	81.7	38.0	54.8
30 m (3 m/s)	60.7	110.4 (18.4)	84.6	30.8	37.2	90.4	86.1	39.4	56.7
40 m (4 m/s)	63.5	115.1 (19.2)	84.6	35.3	41.7	92.3	90.9	44.2	55.3
50 m (5 m/s)	62.2	82.9 (13.8)	84.6	34.6	37.2	92.3	89.4	41.8	55.3

Table 1: FCM Adam — sweep over lr (624 episodes per value, $r_s=50\text{ m}$, $\Delta x_{\max}=50\text{ m}$). “Step (min)” reports the average steps to success and, in parentheses, the corresponding minutes (1 step = 10 s). In green the optimal value $lr=40\text{ m}$ (global SR 63.5%); in bold the best per column.

As lr increases, both the success rate (up to $lr=40\text{ m}$, 63.5%) and the velocity grow: the average steps to success drop from 174 to 83. Beyond $lr=40\text{ m}$ the step becomes so long as to sometimes **overshoot** the source, and the SR drops slightly ($lr=50\text{ m}$: faster but 62.2%). The method’s structural limit emerges from the per-scenario profile: in advected-plume environments (V1, V2) the SR stays stuck at 30–42%. The reason is that there the *local* gradient points along the current and not toward the source: no local momentum can compensate for the absence of global episode memory and hydrodynamic context information. FCM Adam, while gaining a lot over pure FCM, therefore saturates around 63%.

3.2 Adaptive PPO (variable-velocity chain)

Action space. To make velocity a choice of the agent, the action space goes from 8 directions (fixed step) to Discrete($8 \times K$) with $K = 5$: each action *jointly* encodes one of the 8 directions and one of the 5 velocity levels $\{0.2, 0.4, \dots, 1.0\} \cdot v_{\max}$. At each step the agent thus chooses not only *where* to go but also *how hard* to push. The optimal velocity is not imposed by a rule: it emerges **implicitly** from return optimization — the agent could learn, for example, that slowing down near the source or against the current is worthwhile.

Why a warm-start chain. Going directly from a fixed-step model (1 m/s) to one

with free velocity up to 5 m/s is a difficult jump: the action space quintuples (from 8 to 40 actions) and the episode dynamics change. Training is therefore organized as a **chain**, so that each increment is a small adjustment on an already competent policy rather than learning from scratch:

- the first stage ($v_{\max}=1$) starts from the fixed-step *minimal* model, **transplants its body** (the already learned representations) and expands the action head from 8 to 40 outputs; the agent learns to modulate velocity within the same 1 m/s cap;
- each subsequent stage ($v_{\max}=2, \dots, 5$) restarts from the previous one (fine-tuning) and progressively raises the velocity cap.

Each stage is evaluated on 1560 episodes.

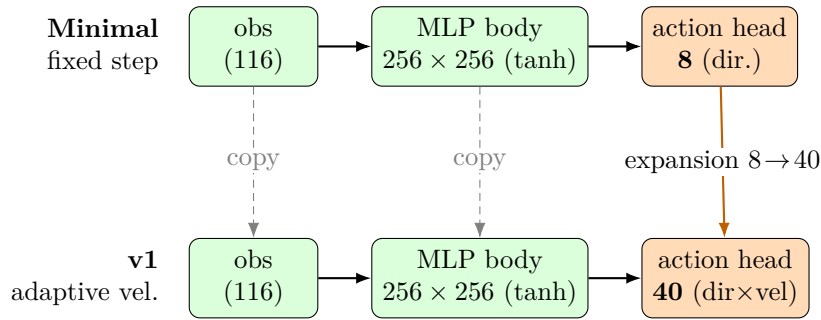


Figure 3: Warm-start of the first stage (*minimal* $\rightarrow$ v1). The network **body** (116-dim input and 256×256 MLP) is **copied** unchanged (green); only the **action head** is **expanded** from 8 to 40 outputs (orange): each direction is replicated across the 5 velocity levels, with an initial bias toward $v_{\max}$ (at start-up the policy matches the fixed-step one).

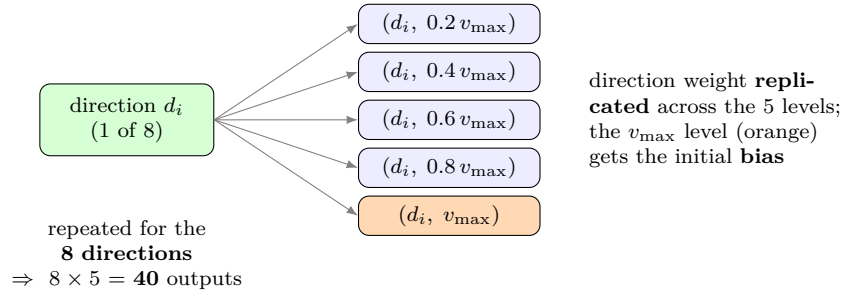


Figure 4: Focus on the action head (expansion $8 \rightarrow 40$). Each of the 8 directions of the *minimal* head is split into the 5 velocity levels (d_i, v_k): the weights are replicated and the $v_{\max}$ level is favored at start-up, so the policy begins equivalent to the fixed-step one.

$v_{\max}$	SR%	Step (min)	V0	V1	V2	V3	Q1/4	Q1/2	Q3/4
1 m/s	97.1	119.6 (19.9)	99.5	100.0	89.7	99.2	100.0	95.0	96.3
2 m/s	99.7	69.5 (11.6)	100.0	99.5	99.2	100.0	100.0	99.0	100.0
3 m/s	99.9	46.9 (7.8)	100.0	100.0	99.7	100.0	100.0	99.8	100.0
4 m/s	99.9	38.2 (6.4)	100.0	100.0	99.5	100.0	100.0	100.0	99.6
5 m/s	99.9	27.8 (4.6)	100.0	100.0	99.7	100.0	100.0	100.0	99.8

Table 2: Variable-velocity PPO chain (1560 episodes per stage, $K=5$). “Step (min)” reports the average steps to success and, in parentheses, the corresponding minutes (1 step = 10 s). As $v_{\max}$ increases, SR and velocity grow (average steps from ~ 20 to ~ 5 min).

Already the first stage ($v_{\max}=1$, velocity-equivalent to the fixed-step model) reaches 97.1%. Raising $v_{\max}$ produces two joint benefits: the SR rises to 99.7–99.9% and the average steps to success collapse from 120 to 28 ($\sim 5\times$ faster). The largest gain is on **V2**, the strong-current scenario: the agent’s net velocity is $v_{\text{agent}} - v_{\text{current}}$, and with more velocity available the agent more easily beats the adverse current, going from 89.7% to 99.7%. From $v_{\max}=3$ on, performance is essentially saturated ($\approx 99.9\%$): the extra velocity margin only reduces the time to success, not the reliability.

Mean velocity over time

For each stage of the chain the mean velocity (over all runs) over time is reported. On average the agent stays close to $v_{\max}$ at every stage; the modulation (the dips of the mean and the band width) is **not monotonic** in $v_{\max}$: it is **maximal at the intermediate stages** ($v_{\max}=2, 3$) and **almost absent at the extremes**. At $v_{\max}=1$ the full step (10 m) is much smaller than the success radius (50 m): there is no risk of overshooting the source, going at maximum is almost always safe and the mean stays flat. At $v_{\max}=5$ the full step (50 m) equals the radius and, thanks to the capture along the travelled segment, a full stride hits the target anyway: again the maximum is worthwhile. At intermediate values, instead, the step becomes comparable with the approach scale and the corrective slow-downs become more evident.

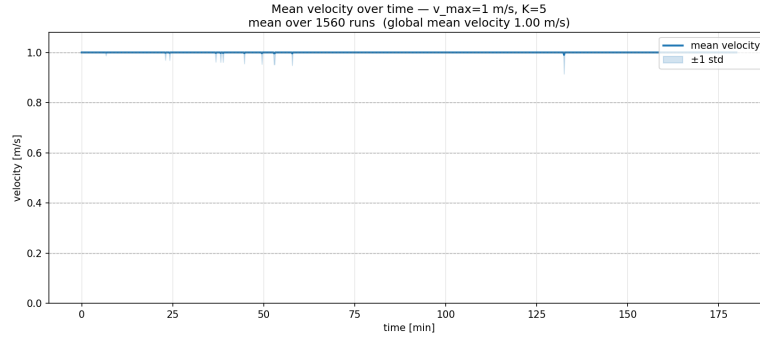
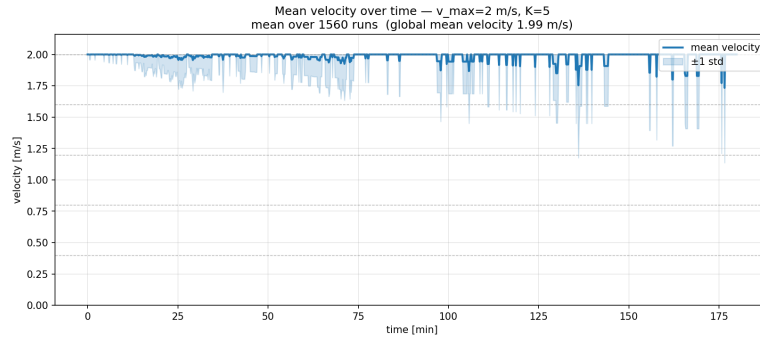
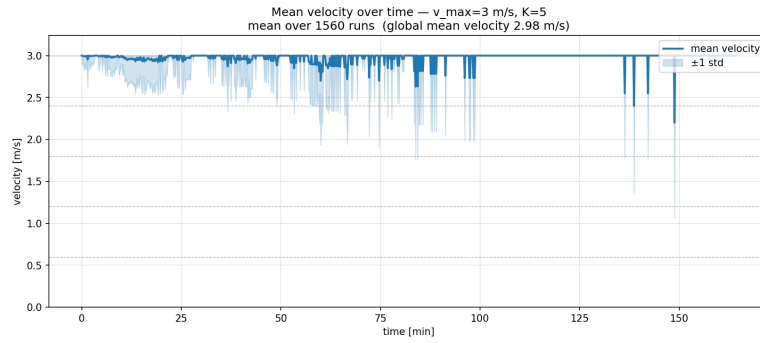
(a) $v_{\max}=1 \text{ m/s}$ (b) $v_{\max}=2 \text{ m/s}$ (c) $v_{\max}=3 \text{ m/s}$

Figure 5: Adaptive PPO: mean velocity (over all runs) over time, stages $v_{\max}=1, 2, 3$ (continued in the next figure). On average the agent stays close to $v_{\max}$ but modulates the step with corrective slow-downs, more evident at the intermediate stages ($v_{\max}=2, 3$) and almost absent at the extremes ($v_{\max}=1$ and 5).

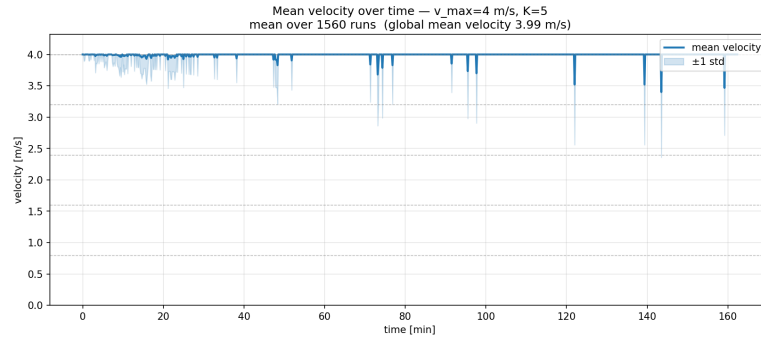
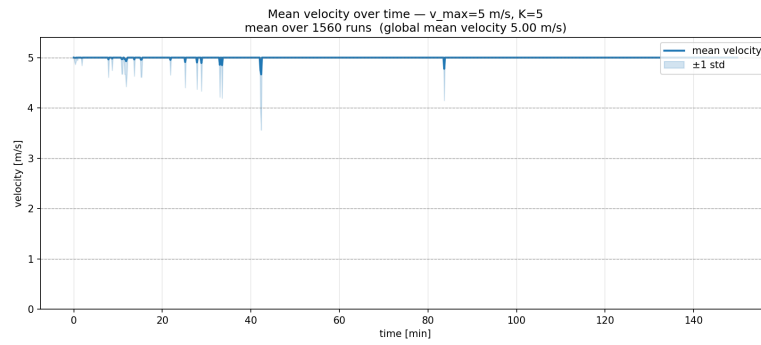
(a) $v_{\max}=4$ m/s(b) $v_{\max}=5$ m/s

Figure 6: Adaptive PPO: mean velocity over time, stages $v_{\max}=4, 5$ (continued from the previous figure).

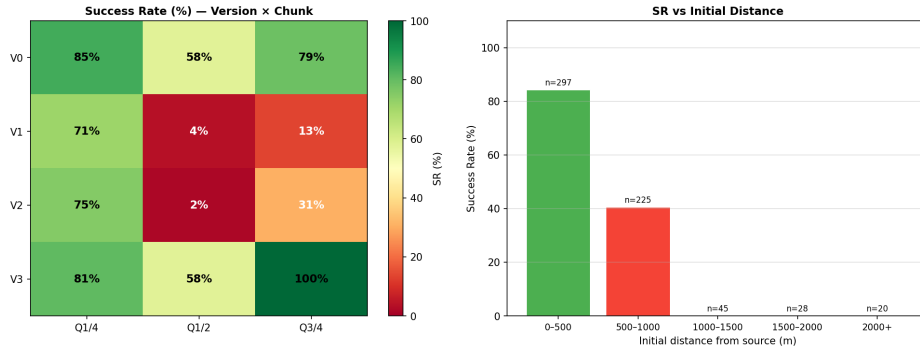
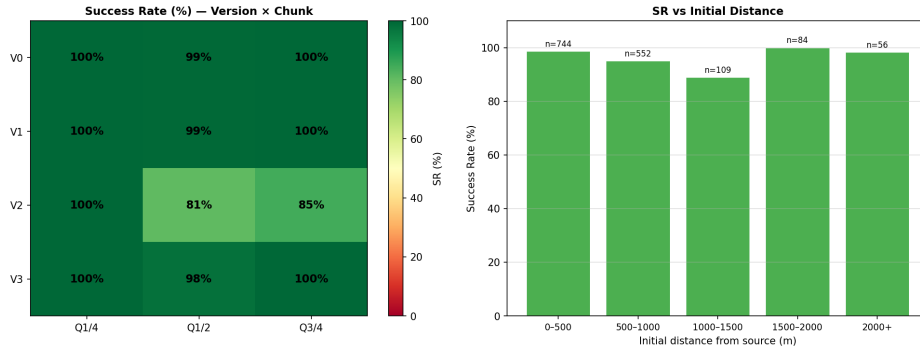
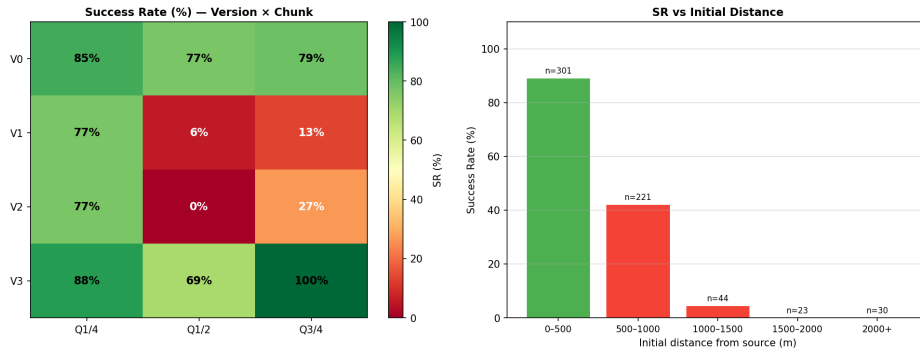
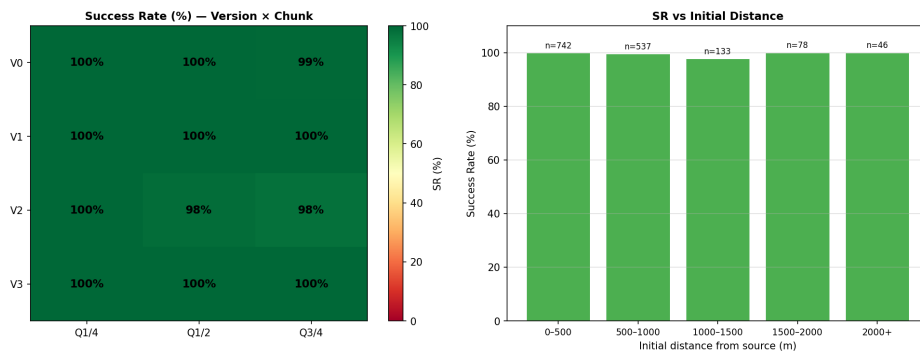
3.3 Quantitative FCM–PPO comparison

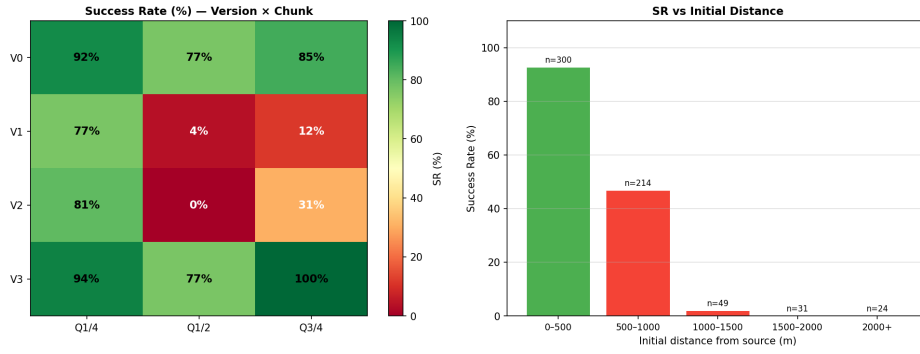
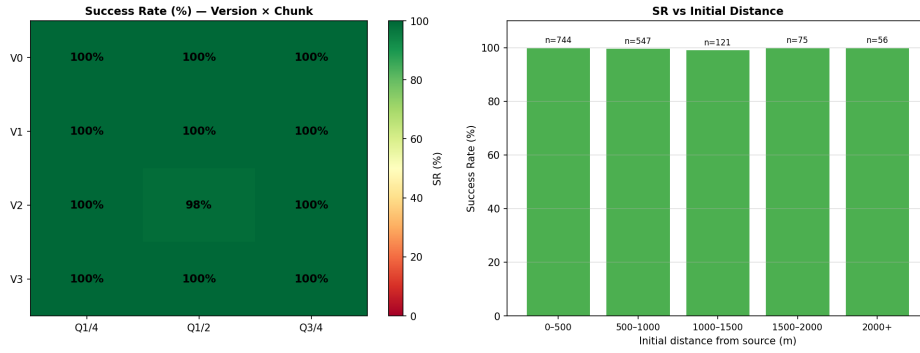
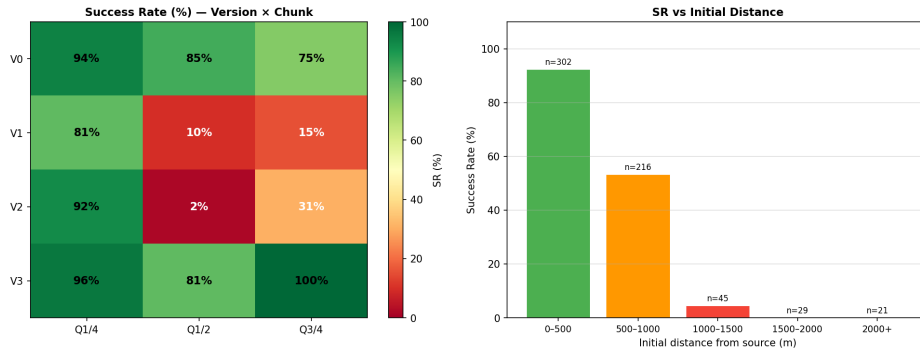
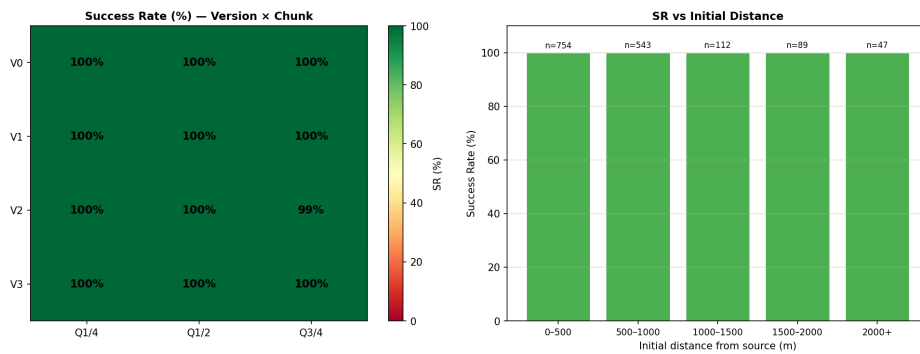
Vel. / method	SR%	V0	V1	V2	V3	Q1/4	Q1/2	Q3/4
FCM ($lr=10$)	54.6	73.7	29.5	35.9	79.5	77.9	30.3	55.8
PPO ($v_{\max}=1$)	97.1 +42.5	99.5 +25.8	100.0 +70.5	89.7 +53.8	99.2 +19.7	100.0 +22.1	95.0 +64.7	96.3 +40.6
FCM ($lr=20$)	58.2	80.1	32.1	34.6	85.9	81.7	38.0	54.8
PPO ($v_{\max}=2$)	99.7 +41.5	100.0 +19.9	99.5 +67.4	99.2 +64.6	100.0 +14.1	100.0 +18.3	99.0 +61.1	100.0 +45.2
FCM ($lr=30$)	60.7	84.6	30.8	37.2	90.4	86.1	39.4	56.7
PPO ($v_{\max}=3$)	99.9 +39.2	100.0 +15.4	100.0 +69.2	99.7 +62.6	100.0 +9.6	100.0 +13.9	99.8 +60.4	100.0 +43.3
FCM ($lr=40$)	63.5	84.6	35.3	41.7	92.3	90.9	44.2	55.3
PPO ($v_{\max}=4$)	99.9 +36.4	100.0 +15.4	100.0 +64.7	99.5 +57.8	100.0 +7.7	100.0 +9.1	100.0 +55.8	99.6 +44.3
FCM ($lr=50$)	62.2	84.6	34.6	37.2	92.3	89.4	41.8	55.3
PPO ($v_{\max}=5$)	99.9 +37.8	100.0 +15.4	100.0 +65.4	99.7 +62.6	100.0 +7.7	100.0 +10.6	100.0 +58.2	99.8 +44.5

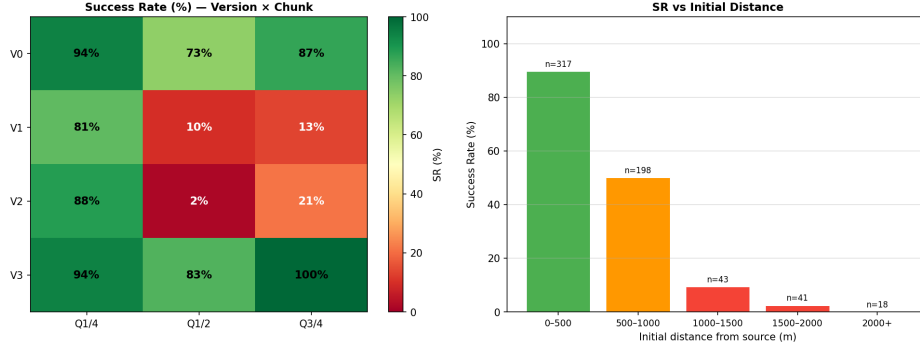
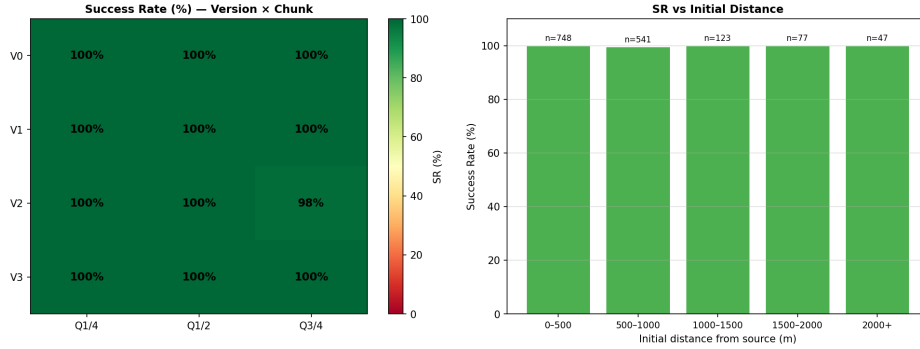
Table 3: Adaptive FCM vs. adaptive PPO at equal maximum velocity (FCM: $lr/\Delta t$; PPO: $v_{\max}$), by wind scenario (V) and time chunk (Q). Under each PPO value, in green, the **surplus** over FCM (percentage points). PPO dominates at every velocity level and on every scenario.

At equal velocity, PPO outperforms FCM by a wide and systematic margin (+36–42 points of global SR). The gap is **maximal** precisely in the most adverse scenarios (V1, V2): there *local* gradient ascent is misled by the advected plume, while the learned policy exploits episode memory and hydrodynamic context so as not to lose the source.

The per-scenario comparison (wind V, chunk Q) is reported **at equal velocity** for each level: in each figure FCM (top) and PPO (bottom) at the same maximum velocity.

(a) FCM Adam ($lr=10$ m).(b) Adaptive PPO ($v_{\max}=1$).Figure 7: Success rate per scenario — **velocity 1 m/s** (FCM top, PPO bottom).(a) FCM Adam ($lr=20$ m).(b) Adaptive PPO ($v_{\max}=2$).Figure 8: Success rate per scenario — **velocity 2 m/s** (FCM top, PPO bottom).

(a) FCM Adam ($lr=30$ m).(b) Adaptive PPO ($v_{\max}=3$).Figure 9: Success rate per scenario — **velocity 3 m/s** (FCM top, PPO bottom).(a) FCM Adam ($lr=40$ m, *best*).(b) Adaptive PPO ($v_{\max}=4$).Figure 10: Success rate per scenario — **velocity 4 m/s** (FCM top, PPO bottom).

(a) FCM Adam ($lr=50$ m).(b) Adaptive PPO ($v_{\max}=5$).Figure 11: Success rate per scenario — **velocity 5 m/s** (FCM top, PPO bottom).

4 Modulation Test: 10 m Success Radius

4.1 Motivation

The results of the variable-velocity chain show that, on average, the agent stays close to $v_{\max}$. A question then arises: does the agent *really* know how to slow down, or is it always better off going at maximum? With a 50 m success radius the question is ill-posed: the radius (50 m) is as large as a full step (50 m at $v_{\max}=5$), so the agent can **swoop** onto the source at full velocity and still hit the 50 m ball, without any need to slow down.

To make the task more stringent, the **success radius is shrunk to 10 m**. Now a full 50 m step **overshoots** a 10 m target, and hitting such a narrow target is no longer trivial. The question is then: to succeed, does the agent learn to *slow down* on approach, or does it adopt another strategy? In either case the expected price is a larger number of steps.

4.2 Setup

The model (ppo_20260630_090846) is obtained by fine-tuning from the $v_{\max}=5$ stage, changing only the success radius ($50 \rightarrow 10$ m), with a reduced stagnation threshold, $lr = 10^{-4}$, curriculum $\alpha = 0.8$, for 2 M timesteps. The evaluation is on 1560 episodes, identical to the other stages.

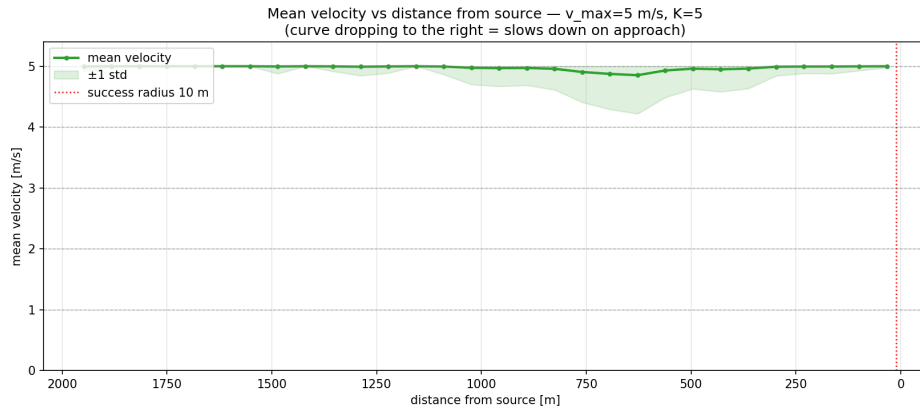
4.3 Results

Radius	SR%	Step (min)	V0	V1	V2	V3	Q1/4	Q1/2	Q3/4
$v_{\max}=5$, 50 m	99.9	27.8 (4.6)	100.0	100.0	99.7	100.0	100.0	100.0	99.8
$v_{\max}=5$, 10 m	99.2 -0.7	54.8 (9.1) +27.0	100.0 0.0	98.5 -1.5	98.5 -1.2	100.0 0.0	98.7 -1.3	99.4 -0.6	99.6 -0.2

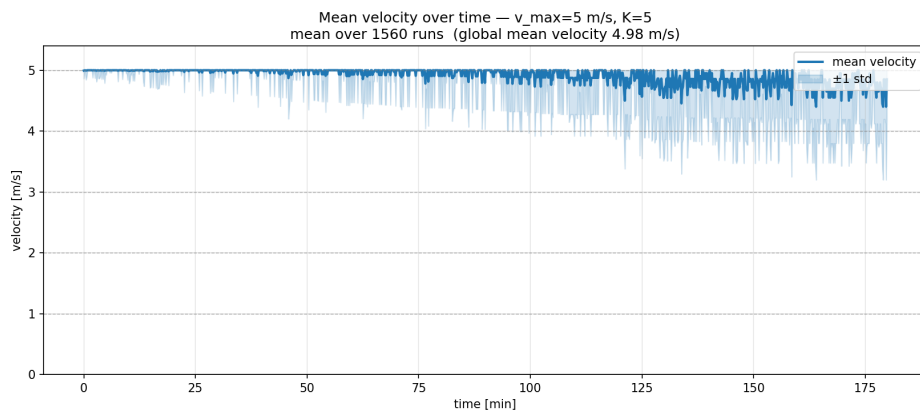
Table 4: Effect of shrinking the success radius ($50 \rightarrow 10$ m) on the $v_{\max}=5$ stage (1560 episodes). Under each 10 m value, the delta with respect to the 50 m configuration (in red the SR drops; in gray the — expected — increase in steps). The success rate stays essentially unchanged (-0.7), but the average steps to success **double** (from ~ 5 to ~ 9 min).

With a target five times smaller, the agent still reaches it almost always (99.2%), but **takes about twice the steps** ($27.8 \rightarrow 54.8$, i.e. $4.6 \rightarrow 9.1$ min). The interesting fact is *where* this extra time goes: **not** into slowing down. The mean velocity in fact stays close to the maximum (4.98 m/s), practically identical to the 50 m configuration.

The reason lies in the capture criterion: success is evaluated on the **travelled segment** (previous $\rightarrow$ current position), not only on the endpoint. A full 50 m step whose trajectory passes within 10 m of the source already counts as a success; the agent is therefore **not forced to brake** — it just needs to aim a stride that grazes the target. Hitting a 10 m target with 50 m steps, however, requires more maneuvers: the agent **overshoots and re-approaches** repeatedly around the source (Fig. 13), accumulating steps at nearly full velocity. The cost of the smaller target is therefore paid in **path length**, not in velocity: the test shows that the policy can *maneuver* finely around a narrow target, more than it shows a systematic slow-down.

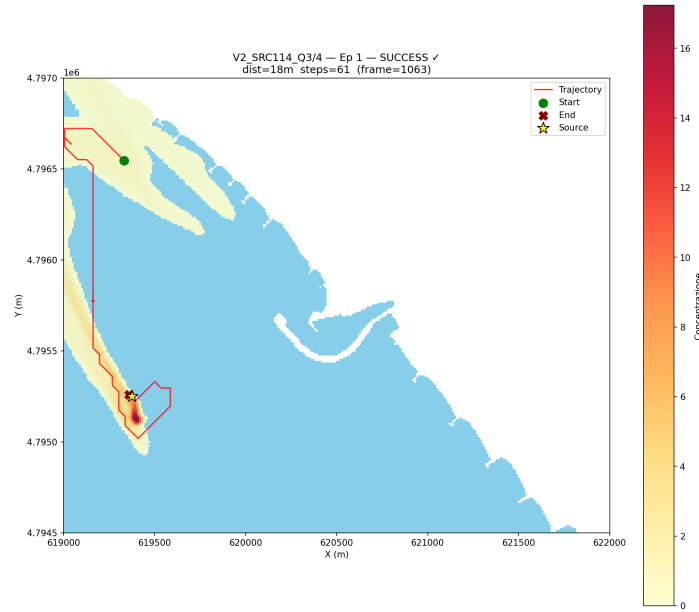


(a) Mean velocity as a function of distance from the source (inverted axis: far on the left, near on the right; red line = success radius).

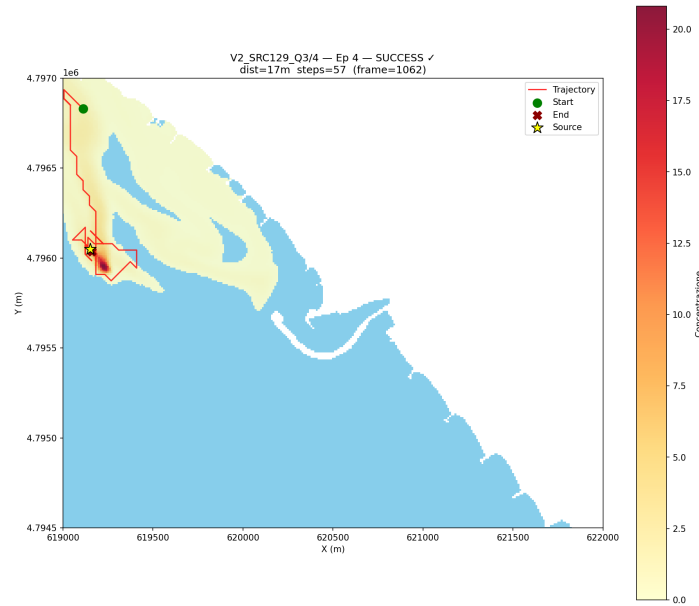


(b) Mean velocity over time.

Figure 12: 10 m radius experiment: the mean velocity stays **close to the maximum** at every distance from the source (a). The apparent downward trend over time (b) is mostly a *selection* effect: at long times only the hard runs survive (V2 against the current), which travel more slowly; it is not a systematic slow-down on approach.



(a) “Clean” capture with a single loop around the source (35 steps).



(b) Intense maneuvering: repeated overshoots and re-approaches (139 steps).

Figure 13: r10 inference trajectories (V2 scenarios, successful outcomes): the agent descends along the plume at full velocity and then **maneuvers / overshoots repeatedly** around the source to pass a stride within 10 m. The cost is a longer path (more steps), not a lower velocity.

5 Single vs. Double Ring Formation

5.1 Motivation and comparison protocol

The double ring adds to the local ring of 8 sensors at $r_s = 20$ m a **second ring** at $r_{s2} = 50$ m, to give the agent information about the plume at a wider spatial scale.

The question is: does this second ring **help**, and **at which velocities**?

In the previous report the double-ring model was trained *from scratch* and showed an SR drop of -3.9 points: that drop, however, **confounded** the sensor’s effect with the greater difficulty of optimizing a larger network from scratch.

For a **fair** comparison **at every velocity level**, the double ring is obtained by **warm-start from the corresponding single-ring model**: for each $v_{\max} \in \{1, \dots, 5\}$, the model `dualcorona_vX` starts from `vmax_X` (same velocity), expanding the input layer from 116 to 196 dimensions and **zeroing** the ring-2 columns. At start-up the network is therefore *identical* to the single ring (it ignores ring-2) and learns during fine-tuning whether and how to use it: any difference is thus attributable to the sensor alone, not to the protocol. Each model is evaluated on 1560 episodes; the following figure illustrates the warm-start (example at $v_{\max}=1$).

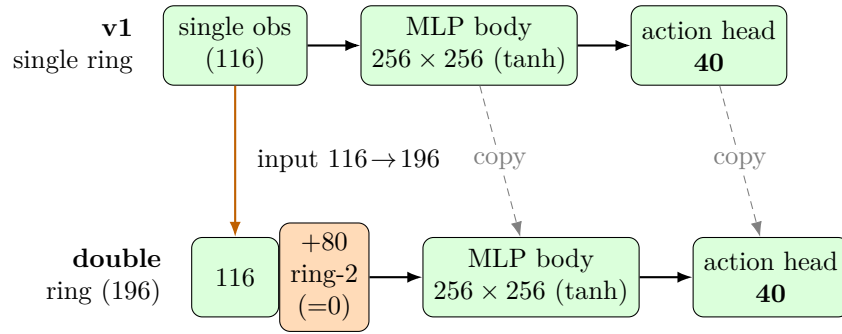


Figure 14: Warm-start of the double ring (example at $v_{\max}=1$: $v1 \rightarrow$ double ring; it is analogous at every level, from the respective `vmax_X`). **Body** and **action head** are **copied** unchanged (green); only the **input layer** is widened from 116 to 196 (orange), adding 80 columns for ring-2 with **weights initialized to zero**. At start-up the network is identical to $v1$ and ignores ring-2; fine-tuning decides whether and how to use it.

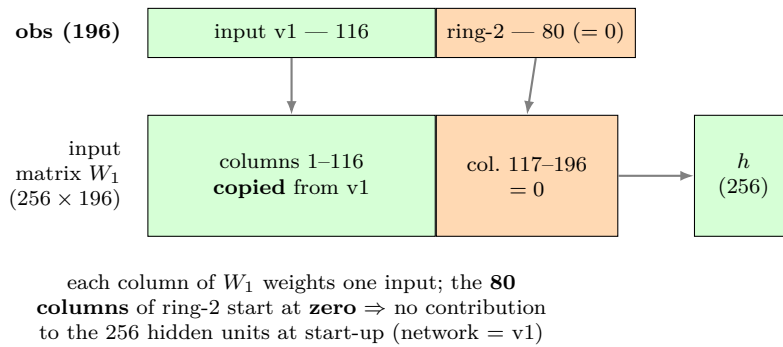


Figure 15: Focus on widening the input layer ($116 \rightarrow 196$). The first-layer matrix gains 80 columns (one per ring-2 sensor), **initialized to zero**: at start-up ring-2 does not affect the hidden units and the network behaves exactly like $v1$.

5.2 Observation architecture

Component	Single	Double
Current concentration	1	1
Concentration memory (9 steps)	9	9
Displacement memory (9 steps)	18	18
Ring-1 sensors (8 dir.)	8	8
Ring-2 sensors (8 dir.)	–	8
Ring-1 history memory (9×8)	72	72
Ring-2 history memory (9×8)	–	72
Wind (2D) + Current (2D)	4	4
Positional information	4	4
Total	116	196

Table 5: Observation space: single (116) vs. double ring (196).

5.3 Results: comparison at each $v_{\max}$

The main result is the overall comparison, single vs. double ring at each velocity level (1560 episodes per cell).

$v_{\max}$	SR single	SR double	Step single	Step double	Δstep
1	97.1%	95.8%	119.6 (19.9 min)	97.8 (16.3 min)	–18.2%
2	99.7%	99.7%	69.5 (11.6 min)	49.8 (8.3 min)	–28.3%
3	99.9%	100.0%	46.9 (7.8 min)	31.3 (5.2 min)	–33.2%
4	99.9%	99.9%	38.2 (6.4 min)	25.3 (4.2 min)	–33.6%
5	99.9%	100.0%	27.8 (4.6 min)	20.9 (3.5 min)	–24.8%

Table 6: Single vs. double ring at each $v_{\max}$ (each double ring warm-started from its single-ring counterpart; 1560 episodes per cell). “Step” = average steps to success (minutes in parentheses). Δstep = relative change in steps (green = double faster).

Two facts emerge, which **overturn** the reading of the previous report (based on $v_{\max}=1$ only, where the second ring seemed to “trade robustness for speed without improving the SR”):

- **The SR cost is confined to $v_{\max}=1$:** the –1.3 penalty does not recur at the higher levels — at v2–v5 the double ring’s SR is *equal to or higher* than the single’s (99.7–100% vs 99.7–99.9%).
- **The step gain does not thin out: it grows.** From –18% (v1) it rises to a peak of $\sim -34\%$ (v3–v4), then reduces to –25% (v5, due to a floor effect: the single is already at ~ 28 steps, close to the geometric minimum).

The reason is that the value of the “far view” (50 m ring) **scales with the step length**: at high $v_{\max}$ the strides are large (up to 50 m) and a badly aimed stride costs an *overshoot* with the related corrections; ring-2, seeing at 50 m, aims the long strides better and avoids exactly those corrections that slow the single down. The practical consequence: **from intermediate velocities up, the double ring is a net advantage** — same reliability and 25–34% fewer steps. Only the base level, $v_{\max}=1$, remains to be understood, the only one where the double ring costs something.

A close look at $v_{\max}=1$

At $v_{\max}=1$ the per-scenario detail (wind V and chunk Q):

Metric	Single (116)	Double (196)
Global SR	97.1%	95.8% (−1.3)
Avg steps (succ.)	119.6 (~19.9 min)	97.8 (~16.3 min)
V0	99.5%	99.7% (+0.2)
V1	100.0%	98.7% (−1.3)
V2	89.7%	86.2% (−3.5)
V3	99.2%	98.7% (−0.5)
Q1/4	100.0%	100.0% 0.0
Q1/2	95.0%	97.1% (+2.1)
Q3/4	96.3%	90.4% (−5.9)

Table 7: $v_{\max}=1$: single (vmax_1) vs. double ring warm-start (dualcorona_v1), 1560 episodes each.

With the warm-start the SR gap **reduces** from −3.9 (from scratch) to −1.3, confirming that part of the original penalty was optimization difficulty, not harmful information. The residual −1.3 is concentrated in **V2** (89.7 → 86.2) and **Q3/4** (96.3 → 90.4); the next section isolates its origin.

5.4 Where the double ring loses: differential failure analysis

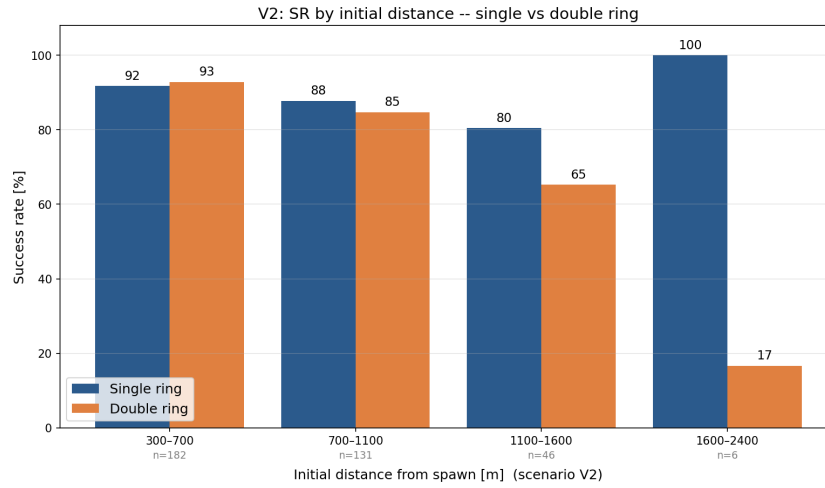
Staying on $v_{\max}=1$ — the only level with an SR cost — by pairing the 1560 episodes by (source, wind, chunk, episode) one isolates the cases where **the single succeeds and the double fails**:

Paired outcome	No. episodes
Both success	1463
Both failure	13
Single OK, double FAILS	52
Single fails, double OK	32
Net	−20 (= −1.3%)

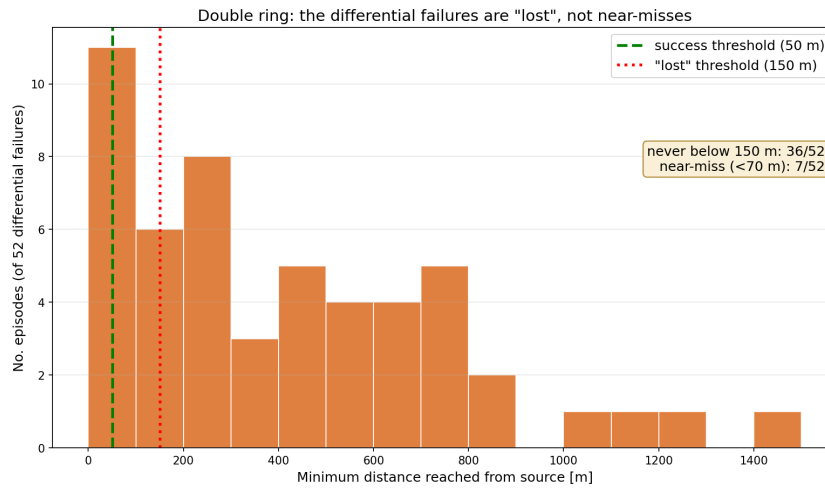
Table 8: Matrix of paired outcomes, single vs. double ring.

The 52 differential failures have a very sharp signature:

- they are **all timeouts** (no domain exits);
- **81% on wind V2** (42/52), in the sustained-current chunks (Q3/4, Q1/2);
- on **far spawns**: average initial distance 1070 m versus 690 m of the general average;
- **not near-misses**: 36/52 never drop below 150 m from the source (median minimum distance 322 m) — the agent **gets lost**, it does not “fail to brake”.



(a) SR by initial distance (V2).



(b) Minimum distance reached over the 52 failures.

Figure 16: Top: in V2 the double ring is equal/better on near spawns and collapses on far ones. Bottom: the differential failures are “lost” (never below 150 m), not stalled approaches.

The collapse is entirely in the “far + strong current” regime: in V2 the SR by initial-distance band goes from +1.1 (300–700 m, double slightly better) to −15.2 (1100–1600 m) down to −83.3 (beyond 1600 m).

The opposite case: where the double ring wins

For completeness, the **32 cases where the single fails and the double succeeds** must also be analyzed. They too are **all timeouts** of the single and **concentrated in V2** (28/32), but with a *complementary* signature compared with the 52 failures:

- the dominant chunk is **Q1/2** (25/32), not Q3/4;
- the spawns are at **moderate** distance (average 749 m, versus the 1070 m of the cases where the double loses);
- the single fails there in a mixed way — 11/32 near-miss (reaches < 70 m and

stalls) and 14/32 “lost” — while the double recovers them, but **at a high price**: 258 average steps (~ 43 min).

It emerges that the double ring **is not uniformly worse: it redistributes** the successes along the temporal axis. On the **Q1/2** chunks (moderate-distance spawns) the 50 m signal helps hold the course and is in **surplus** (25 recoveries against 14 losses); on the **Q3/4** chunks (farther spawns, more sustained current) the same signal misleads and is in **deficit** (7 recoveries against 38 losses). The overall balance is -20 : a **shift** of the success frontier, not a generalized regression.

5.5 Interpretation

The thread that holds all the results together is a single one: ring-2 senses the field **farther away** (50 m), and the value of this “far view” depends on *how far the agent is about to move*, i.e. on the step length.

Why the speed advantage grows with $v_{\max}$. At high $v_{\max}$ the strides are large (up to 50 m): a badly aimed stride overshoots the course and costs corrections. Ring-2, seeing at 50 m, is exactly “one step ahead” and allows aiming the long strides well, avoiding the *overshoots* that slow the single down. At low $v_{\max}$ the step is small (20 m): the far signal is less decisive, and indeed the gain is smaller (-18% at $v1$ versus $\sim -34\%$ at $v3$ – $v4$). The slight drop at $v5$ is a floor effect: the single is already close to the step minimum.

Why the reliability cost exists only at $v_{\max}=1$. With small steps the agent stays immersed in the field longer, and in **V2** (strong current, strongly advected plume) the 50 m signal can point *along the current* rather than toward the source: on far spawns the agent trusts the distant signal, takes the wrong direction and times out. It is the same mechanism that gives speed, flipped in the “far + strong current” regime — but only at $v1$, where the small steps leave time to get lost. At the higher velocities the agent crosses that regime too fast for the error to accumulate, and the cost disappears.

Why the agent does not ignore ring-2. “Ignoring” amounts to input weights exactly null, a solution that gradient descent does not reach (even starting from zero as in the warm-start, the gradients move those weights). Above all, **using it increases the reward**: the `step_penalty` rewards arriving sooner, and ring-2 gives trajectories more direct; the optimizer therefore *chooses* to use it.

In summary. The second ring is a **net advantage** from $v_{\max}=2$: same reliability and 25–34% fewer steps. Only at $v_{\max}=1$ is a small robustness/speed trade-off observed (-1.3 of SR for -18% of steps), confined to far spawns in strong current. The previous report’s reading — “the double ring does not improve the SR, it trades robustness for speed” — was an artifact of having looked at $v_{\max}=1$ *only*.

5.6 Agent occupancy heatmaps

For a *visual* confirmation of the numerical results, some **occupancy heatmaps** are reported: given a scenario (source, wind, chunk), 10 episodes are run and the agent’s **presence density** over the domain is accumulated (“sports heatmap” style), on a geographic background (sea, coast, source $\star$). Three agents are compared at the **same maximum velocity**: **FCM** (5 m/s), **PPO single ring** ($v_{\max}=5$) and **PPO double ring** ($v_{\max}=5$), over 5 *hard* scenarios (wind V1/V2, chunk Q1/2 or Q3/4). The color

scale is **common to the three panels**, fixed on the two (comparable) PPO models; the FCM, which oscillates for a long time in the same spots, has much higher peak densities and therefore **saturates**.

Remarks.

- **FCM vs. PPO.** The FCM stays **trapped** in oscillating blobs far from the source and almost always fails (0–5/10), while the PPO forms a trail that **converges on the source** (10/10). It is the visual counterpart of the SR gap (+36–42 points): *local* gradient ascent, with no memory or context, gets lost in the advected plume.
- **Single vs. double ring.** Both reach the source, but the **double ring concentrates the presence in a narrower, more defined corridor** (brighter at equal scale), while the single is more dispersed. It is the visual counterpart of the –34% of steps: the second ring makes the route more direct and repeatable.
- **How to read it.** The brightness indicates the *transit density*, not the velocity: a brighter stretch near the source reflects the **convergence** of all the runs into a common corridor, not a slow-down (the velocity stays $\sim v_{\max}$).

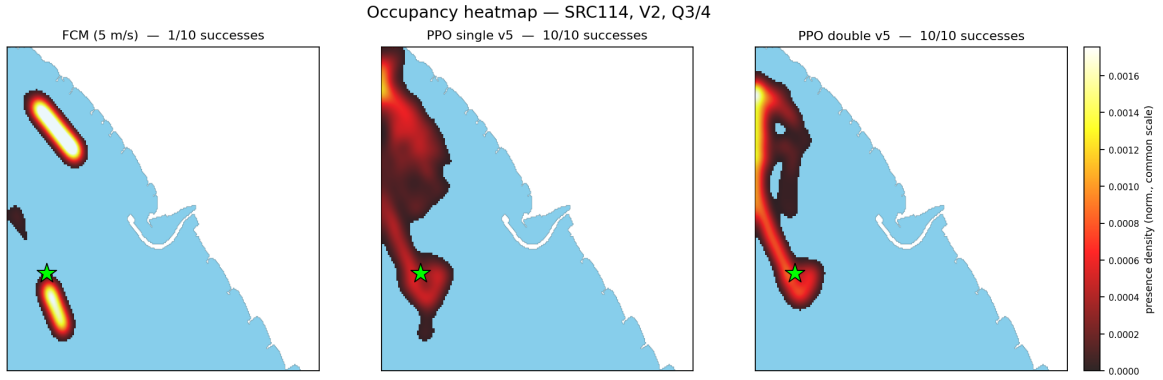


Figure 17: Occupancy heatmap — SRC114, V2, Q3/4. FCM (left) vs. PPO single (center) vs. PPO double ring (right), $v_{\max}=5$, common scale.

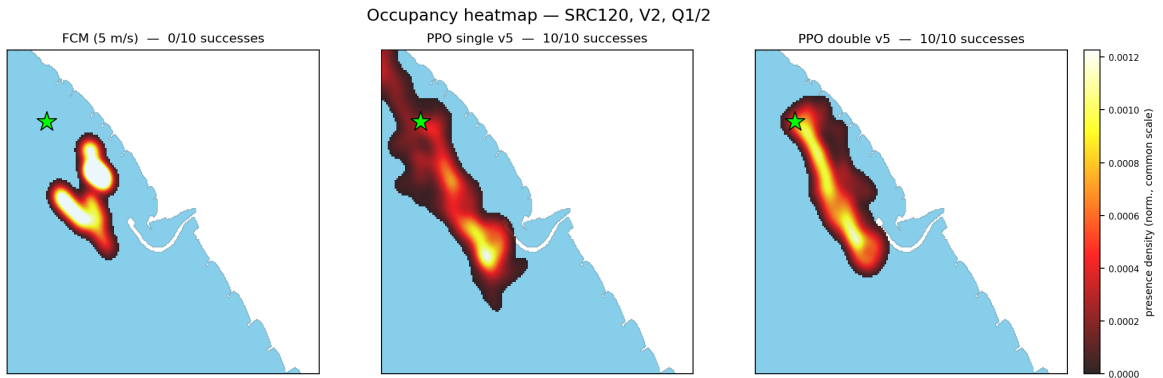


Figure 18: Occupancy heatmap — SRC120, V2, Q1/2.

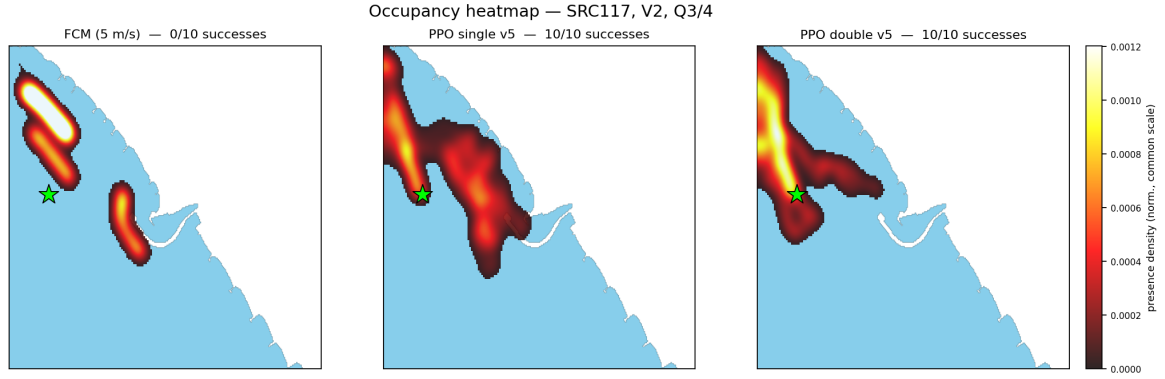


Figure 19: Occupancy heatmap — SRC117, V2, Q3/4.

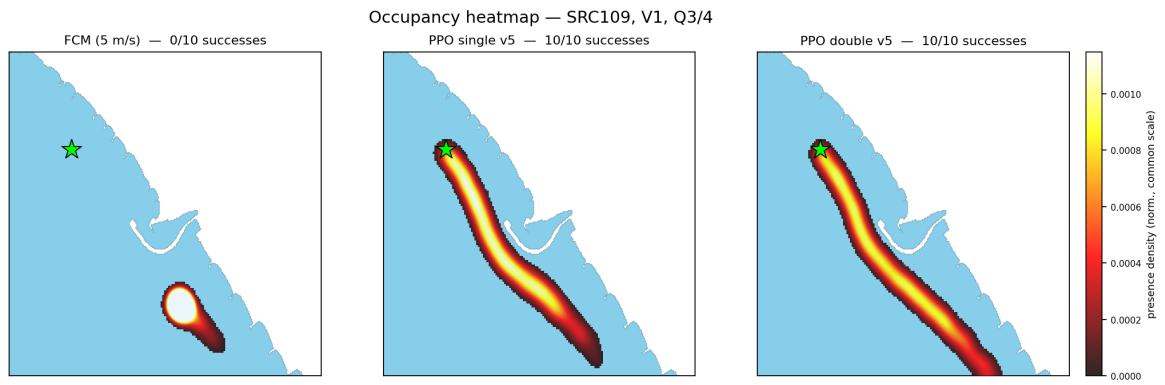


Figure 20: Occupancy heatmap — SRC109, V1, Q3/4.

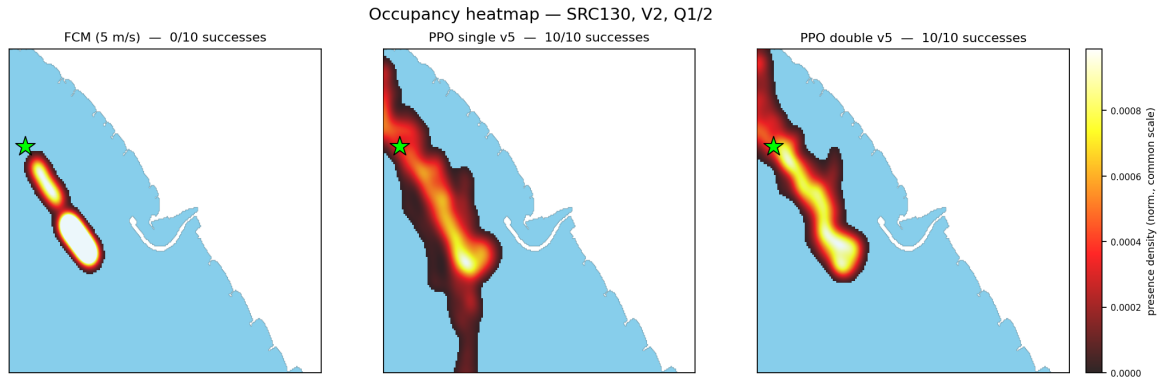


Figure 21: Occupancy heatmap — SRC130, V2, Q1/2.

6 Conclusions

In this report we compared a gradient method (FCM) and a policy learned with PPO for the localization of marine pollutant sources with **adaptive-step** agents, along four axes.

1. **Uniform spawn in $[d_{\min}, d_{\max}]$.** The new metric distributes the start points *uniformly by distance* in a disk adapted to the plume's diffusion (inner edge =

median of the plume distances, with a 300 m floor; outer edge ≤ 3 km). The spawn maps confirm that successes and failures have **no** spatial separation: the outcome depends on the local hydrodynamic conditions, not on the spawn geometry.

2. **Adaptive FCM vs. adaptive PPO.** At equal maximum velocity, PPO **dominates** FCM by +36–42 points of SR. Adam’s momentum brings FCM from 32.9% (fixed step) to $\sim 63\%$, but there it saturates: *local* gradient ascent does not replace the global episode memory and the hydrodynamic context that the learned policy exploits — especially in the adverse scenarios (V1, V2), where the local gradient is misled by the advected plume.
3. **Velocity modulation (10 m radius).** Shrinking the success radius from 50 to 10 m, the agent still succeeds (99.2%) but doubles the steps. The extra time does **not** come from a slow-down (mean velocity unchanged, ~ 4.98 m/s): with the capture on the travelled segment the agent only needs to *graze* the target, so it does not brake but **maneuvers** with a longer path.
4. **Single vs. double ring.** With a fair comparison at *every* $v_{\max}$ (warm-start of the double ring from its single-ring counterpart), the second ring is a **net advantage from** $v_{\max} \geq 2$: same reliability (99.7–100%) and 25–34% fewer steps. The SR cost (−1.3) exists **only** at $v_{\max}=1$, on far spawns in strong current (V2). The advantage grows with velocity because the “far view” (50 m) is worth more when the strides are large.

Summary. The PPO-learned agent clearly outperforms the gradient method, and the ability to **modulate velocity** combined with a **second sensor formation** makes the search faster without sacrificing reliability. The value of the second ring, in particular, is appreciated only by evaluating it over the **whole** velocity range: at $v_{\max}=1$ it would seem harmful, while from $v_{\max} \geq 2$ it is a net advantage (same reliability, up to −34% of steps).